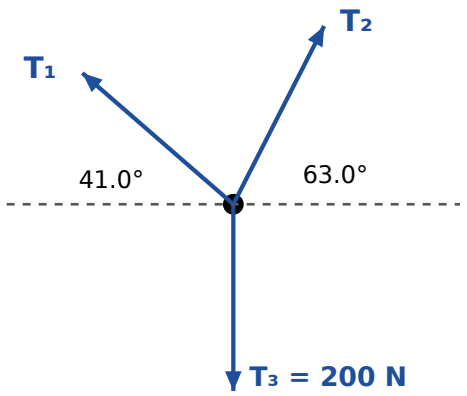
WS Homework 6: Applications of Newton's Laws

Answer sheet with work shown • g = 9.80 m/s² • final answers circled in red

Problem 1 — Tensions in the traffic-light cables

The light hangs in equilibrium, so $T_3 = w$. Then analyze the knot where the three cables meet. Angles 41.0° and 63.0° are measured from the horizontal ceiling.



FBD of the knot

Light: $T_3 - w = 0 \rightarrow T_3 = 200$ N

Knot, x: $T_2 \cos 63.0° - T_1 \cos 41.0° = 0$
Knot, y: $T_1 \sin 41.0° + T_2 \sin 63.0° - T_3 = 0$

From x: $T_1 = T_2 (\cos 63.0° / \cos 41.0°)$
$= T_2 (0.4540 / 0.7547) = 0.6015\, T_2$

Substitute into the y-equation:

$0.6015\, T_2 (0.6561) + T_2 (0.8910) = 200$ N

$T_2 (0.3946 + 0.8910) = 200$ N $\rightarrow$ $T_2 = 200$ N / 1.2856 = 155.6 N

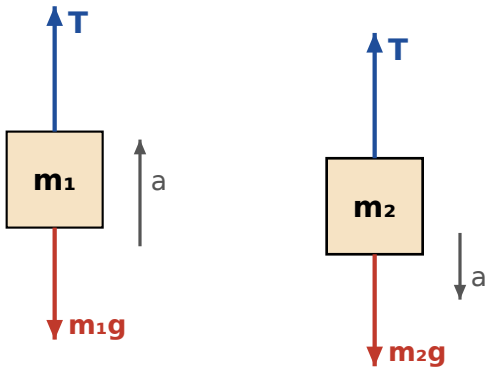
$T_1 = 0.6015$ (155.6 N) = 93.6 N

Check y: 93.6(0.6561) + 155.6(0.8910) = 61.4 + 138.6 = 200 N (checks)

T_1 = 93.6 N T_2 = 156 N T_3 = 200 N

Problem 2 — Atwood's machine

Take $m_2 > m_1$, so m_2 accelerates down and m_1 accelerates up with the same magnitude a. Same tension T throughout (massless string, frictionless massless pulley).



FBDs (gray arrow = direction of a)

(a) m_1: $T - m_1 g = m_1 a$

m_2: $m_2 g - T = m_2 a$

Add the equations (T cancels):

$m_2 g - m_1 g = (m_1 + m_2) a$

$$\mathbf{a = (m_2 - m_1)\, g \,/\, (m_1 + m_2)}$$

(b) Substitute a into $T = m_1(g + a)$:

$T = m_1 g\,[1 + (m_2 - m_1)/(m_1 + m_2)] = m_1 g\,[2m_2/(m_1 + m_2)]$

$$\mathbf{T = 2 m_1 m_2\, g \,/\, (m_1 + m_2)}$$

(c) $m_1 = 2.00$ kg, $m_2 = 4.00$ kg:

$a = (4.00 - 2.00)(9.80) / (2.00 + 4.00) = 19.6 / 6.00 = 3.27$ m/s²

$T = 2(2.00)(4.00)(9.80) / 6.00 = 156.8 / 6.00 = 26.1$ N

Check: $T = m_1(g + a) = 2.00(9.80 + 3.27) = 26.1$ N (checks)

a = 3.27 m/s² (m_2 down, m_1 up) T = 26.1 N

Problem 3 — Two carts on inclines over a pulley

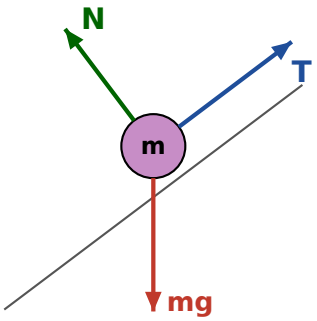
10 kg cart on the 37° incline, 15 kg cart on the 53° incline, no friction.
Only gravity components along the inclines matter (normal forces are perpendicular to motion)..

Compare the pulling components along each incline:

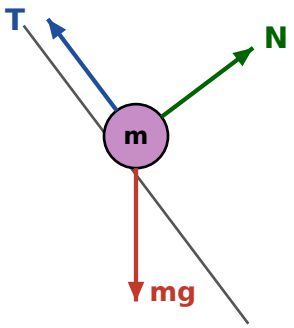
15 kg: $m_2 g \sin 53° = (15)(9.80)(0.7986) = 117.4$ N (down its incline)

10 kg: $m_1 g \sin 37° = (10)(9.80)(0.6018) = 59.0$ N (down its incline)

117.4 N > 59.0 N, so the 15 kg cart slides down and the 10 kg cart is pulled up.



10 kg cart on 37°



15 kg cart on 53°

Newton's 2nd law along each incline (positive = direction of motion):

10 kg: $T - m_1 g \sin 37° = m_1 a$

15 kg: $m_2 g \sin 53° - T = m_2 a$

Add: $m_2 g \sin 53° - m_1 g \sin 37° = (m_1 + m_2) a$

$a = (117.4 \text{ N} - 59.0 \text{ N}) / (10 \text{ kg} + 15 \text{ kg}) = 58.4 \text{ N} / 25 \text{ kg}$

$a = 2.34$ m/s²

Tension from the 10 kg cart: $T = m_1(a + g \sin 37°)$

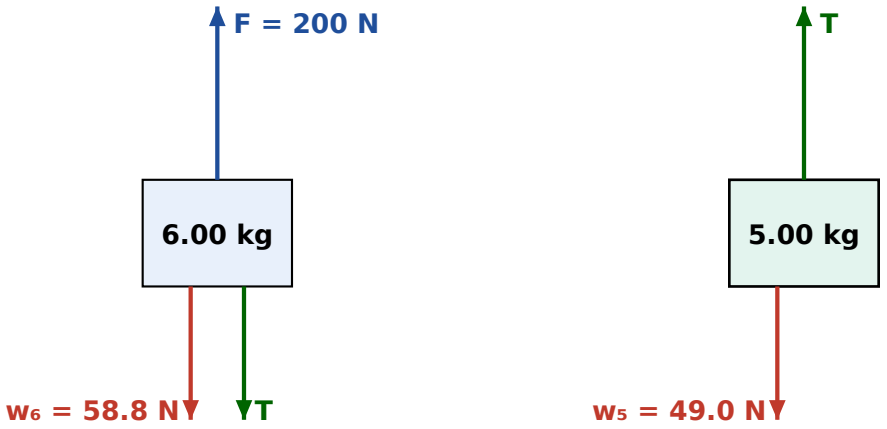
$T = 10(2.34 + 5.90) = 82.4$ N

Check with 15 kg cart: $T = 15(9.80 \cdot 0.7986 - 2.34) = 15(7.83 - 2.34) = 82.4$ N (checks)

$T = 82.4$ N

Problem 4 — Two blocks pulled up by F = 200 N

(a) Free-body diagrams



6.00 kg block **5.00 kg block**

6.00 kg: F (exerted by the person/agent pulling up), T (exerted by the rope, down), $w_6 = m_6 g$ (exerted by Earth, down)

5.00 kg: T (exerted by the rope, up), $w_5 = m_5 g$ (exerted by Earth, down)

(b) Treat both blocks as one system (rope tension is internal and cancels):

$F - (m_6 + m_5)g = (m_6 + m_5)a$

$a = [200 - (11.0)(9.80)] / 11.0 = (200 - 107.8) / 11.0 = 92.2 / 11.0$

$a = 8.38$ m/s² (upward)

(c) Apply Newton's 2nd law to the 5.00 kg block alone:

$T - m_5 g = m_5 a \rightarrow T = m_5(g + a) = 5.00(9.80 + 8.38)$

Check with 6.00 kg block: $F - T - m_6 g = m_6 a \rightarrow 200 - 90.9 - 58.8 = 50.3 = 6.00(8.38)$ (checks)

$T = 90.9$ N